

Breakthrough in optical communication

IBM has succeeded in the final technological hurdle to complete its nanophotonic toolkit. The company claims that its 40Gbps germanium avalanche photodetector is a keynote achievement in on-chip optical communications. IBM now has everything to start integrating photonic communications alongside transistors and make this dream a reality.

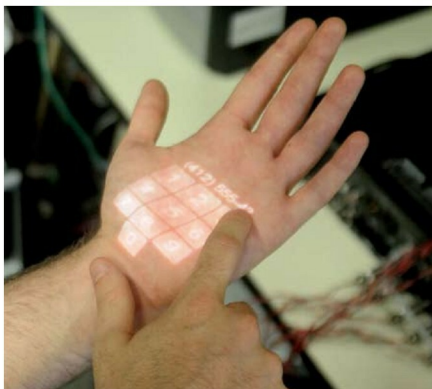
IBM's germanium photodetector multiplies input signals by ten times using the avalanche effect, yet achieves 40 Gbps due to its thin-film construction. It is just 30-nanometres thick, compared to hundreds of nanometres for competing germanium photodetector designs, enabling its 40Gbps speed. Its ultra-thin construction also reduces the normal noise associated with germanium photodetectors by 50 to 70 per cent, thereby resurrecting a technology that was once thought to be too noisy for commercialisation.

The device runs off a 1.5V supply, making it ideal for integration on silicon chips compared to traditional avalanche photodetectors that run off 30V supplies.

New technology for a wireless world

Purdue University researchers have developed a miniature device that converts ultrafast laser pulses into radio-frequency signals. The new device will make communication devices in homes, businesses and cars completely wireless.

The advance could enable all communications, from high-definition television broadcasts to secure computer



Skinput, appropriating the body as an input surface

connections, to be transmitted from a single base station. This base station would be sort of a computer by itself. The central computer would take charge of all the information processing, acting as a single point of contact that interacts with the external world in receiving and sending information.

The pulsing nature of the signals produced by the chip-based spectral shaper reduces the interference that normally plagues radio frequency communications.

Your hand to double as touchpad

With a system named as 'Skinput' developed by Chris Harrison at Carnegie Mel-

lon University in Pittsburgh, and Dan Morris and Desney Tan at Microsoft's research lab in Redmond, USA, you can now have a keypad on your arm.

The Skinput system combines two technologies: the ability to detect the ultra-low-frequency sound produced by tapping the skin with a finger, and the microchip-sized pico projectors found in some cellphones.

The system beams a keyboard or menu onto the user's forearm and hand from a projector housed in an armband. Various locations on the forearm and hand produce characteristic acoustic patterns when tapped. The acoustic detector in the armband contains five piezoelectric cantilevers, each

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